Theoretical Analysis for the Performance of the Tree-Transformer Power Predictor

A. Proof of Theorem 1

We apply the following steps:

Step 1: *Regularity of* $f_k^\star$. By Assumption 1, $f_k^\star$ is L_f-Lipschitz on the compact domain $\mathcal{X} \subset \mathbb{R}^{d_{\text{in}}}$ with $d_{\text{in}} = 2(K+L)$. Since $\mathcal{X}$ is compact and $f_k^\star$ is Lipschitz, then by definition of Hölder classes, it belongs to the Hölder class $\mathcal{C}^{0,1}(\mathcal{X})$ of functions with Hölder smoothness of order $s = 1$ [1, Sect. 4.1].

Step 2: *Application of Yarotsky's theorem.* By [2, Theorem 1], for any scalar-valued function $f_k^\star \in \mathcal{C}^{0,1}([0,1]^{d_{\text{in}}})$ and $\varepsilon \in (0,1)$, there exists a ReLU neural network f_θ with at most $\mathcal{P}$ parameters such that

$$\sup_{x\in\mathcal{X}} |f_{\theta,k}(x) - f_k^\star(x)| \le \varepsilon, \tag{1}$$

$$\mathcal{P} \le Q\,\varepsilon^{-d_{\text{in}}}\big(\ln(1/\varepsilon) + 1\big) \tag{2}$$

where $Q > 0$ depends only on d_{in} and the Lipschitz constant L_f of $f^\star$.

Step 3: *Bounding* ε. From (2), we can write

$$\mathcal{P} \le Q\,\varepsilon^{-d_{\text{in}}}\ln(1/\varepsilon) \tag{3}$$

Let $t = 1/\varepsilon$. The inequality becomes

$$\mathcal{P} \le Q\,t^{d_{\text{in}}}\ln t. \tag{4}$$

We derive a lower bound on t via contradiction. Suppose

$$t < A\left(\frac{\mathcal{P}}{\ln\mathcal{P}}\right)^{1/d_{\text{in}}}, \tag{5}$$

where $A > 0$ is a constant, which implies

$$\begin{aligned} \ln t &< \ln A + \frac{1}{d_{\text{in}}}\ln\mathcal{P} - \frac{1}{d_{\text{in}}}\ln\ln\mathcal{P} \\ &< \ln A + \frac{1}{d_{\text{in}}}\ln\mathcal{P}. \end{aligned} \tag{6}$$

For $\mathcal{P} \ge e$, we have $\ln\mathcal{P} \ge 1$, hence

$$\ln A \le |\ln A|\ln\mathcal{P}, \tag{7}$$

which yields

$$\ln t \le \left(\frac{1}{d_{\text{in}}} + |\ln A|\right)\ln\mathcal{P}. \tag{8}$$

Define

$$Q_1 \triangleq \frac{1}{d_{\text{in}}} + |\ln A|. \tag{9}$$

Then

$$\ln t \le Q_1 \ln\mathcal{P}, \tag{10}$$

and

$$t^{d_{\text{in}}}\ln t \le A^{d_{\text{in}}}\frac{\mathcal{P}}{\ln\mathcal{P}}\cdot Q_1\ln\mathcal{P} = Q_1 A^{d_{\text{in}}}\mathcal{P}. \tag{11}$$

Multiplying by C,

$$Qt^{d_{\text{in}}}\ln t \le QQ_1A^{d_{\text{in}}}\mathcal{P}. \tag{12}$$

Choose $A > 0$ sufficiently small such that

$$QQ_1A^{d_{\text{in}}} < 1. \tag{13}$$

Then

$$Qt^{d_{\text{in}}}\ln t < \mathcal{P}, \tag{14}$$

which contradicts the starting assumption in (3). Therefore, the assumption on t is false, and we conclude that

$$t \ge A\left(\frac{\mathcal{P}}{\ln\mathcal{P}}\right)^{1/d_{\text{in}}}, \tag{15}$$

with $A^{d_{\text{in}}} < 1/(QQ_1)$. Returning to $\varepsilon = 1/t$, we obtain

$$\varepsilon \le Q_3\left(\frac{\mathcal{P}}{\ln\mathcal{P}}\right)^{-1/d_{\text{in}}} \tag{16}$$

for a constant $Q_3 = 1/A$. Therefore,

$$\sup_{x\in\mathcal{X}} |f_{\theta,k}(x) - f_k^\star(x)| \le Q_3\left(\frac{\mathcal{P}}{\ln\mathcal{P}}\right)^{-1/d_{\text{in}}}. \tag{17}$$

Step 4: *Extension to vector-valued outputs.* The optimal mapping $f^\star : \mathcal{X} \to \mathbb{R}^{2K}$ has $2K$ scalar output components $f_1^\star, \dots, f_{2K}^\star$. Since $f^\star$ is L_f-Lipschitz, each component $f_i^\star$ is also L_f-Lipschitz. Therefore, Step 3 applies to each component with the same constant Q_3. The Tree-Transformer decoder produces all $2K$ outputs through a shared decoder network, which outputs all components jointly; thus the bound applies componentwise to each output coordinate:

$$\begin{aligned} &\sup_{x\in\mathcal{X}} |f_{\theta,i}(x) - f_i^\star(x)| \le Q_3\left(\frac{\mathcal{P}}{\ln\mathcal{P}}\right)^{-1/d_{\text{in}}}, \\ &\qquad \forall\, i = 1,\dots,2K. \end{aligned} \tag{18}$$

Taking the maximum over all components and using the definition of the ℓ_∞ norm on $\mathbb{R}^{2K}$:

$$\begin{aligned}\sup_{x\in\mathcal{X}} &\|f_\theta(x) - f^\star(x)\|_\infty \\ &= \max_{i=1,\ldots,2K} \sup_{x\in\mathcal{X}} |f_{\theta,i}(x) - f_i^\star(x)| \\ &\leq Q_3 \left(\frac{\mathcal{P}}{\ln \mathcal{P}}\right)^{-1/d_{\text{in}}}. \end{aligned} \tag{19}$$

Step 5: *Application to mean square error (MSE).* For any vector $v \in \mathbb{R}^{2K}$, the standard inequality between ℓ_2 and ℓ_∞ norms gives:

$$\|v\|^2 \leq 2K\|v\|_\infty^2. \tag{20}$$

Thus,

$$\|f_\theta(x) - f^\star(x)\|^2 \leq 2K \cdot Q_3^2 \left(\frac{\mathcal{P}}{\ln \mathcal{P}}\right)^{-2/d_{\text{in}}}. \tag{21}$$

Taking expectation over X yields

$$\begin{aligned}\mathcal{E}_{\text{approx}} &= \mathbb{E}_X \|f_\theta(X) - f^\star(X)\|^2 \\ &\leq 2K \cdot Q_3^2 \left(\frac{\mathcal{P}}{\ln \mathcal{P}}\right)^{-2/d_{\text{in}}}. \end{aligned} \tag{22}$$

Defining

$$C_{\text{approx}} \triangleq 2K \cdot Q_3^2, \tag{23}$$

as a constant that depends on d_{in} and L_f, we obtain

$$\mathcal{E}_{\text{approx}} \leq C_{\text{approx}} \left(\frac{\mathcal{P}}{\ln \mathcal{P}}\right)^{-2/d_{\text{in}}}. \tag{24}$$

Proof of Theorem 2

To prove Theorem 2, we apply the following steps:

Step 1: *Bound using Rademacher complexity.* The generalization performance (i.e., estimation error) of the learning model is controlled by the Rademacher complexity $\mathcal{R}_N$ [3], [4], which measures how well a model class can fit random noise. Large $\mathcal{R}_N$ means that the model is highly complex and can fit noise well leading to a higher risk of overfitting while small $\mathcal{R}_N$ implies that the model's complexity is controlled, leading to less overfitting and better generalization. In other words, the Rademacher complexity measures the complexity of the model and its capacity to overfit random data, which both affect directly the generalization performance of the model.

Let $\mathcal{F} = \{f_\theta\}$ be the model class induced by the Tree-Transformer (TT) architecture, $\mathcal{L}(\cdot,\cdot)$ be a bounded and Lipschitz loss function in its first argument (predicted power), and let $\mathcal{L} \circ \mathcal{F}$ denote the loss-composed function class:

$$\mathcal{L} \circ \mathcal{F} = \{(x, y) \mapsto \mathcal{L}(f(x), y) : f \in \mathcal{F}\}. \tag{25}$$

Assume that the loss function ℓ is bounded in $[0, 1]$. Then, for any $\delta \in [0, 1]$, with probability at least $1 - \delta$, the following inequality holds for all $f \in \mathcal{F}$ according to [4, Theorem 8]:

$$\mathcal{E}_{\text{est}} \leq \mathcal{R}_N(\mathcal{L} \circ \mathcal{F}) + \sqrt{\frac{8 \ln(2/\delta)}{S}}, \tag{26}$$

where $\mathcal{R}_N(\mathcal{L} \circ \mathcal{F})$ denotes the Rademacher complexity of the loss-composed function class. This bound requires the loss to be bounded in $[0, 1]$; for MSE on normalized outputs this holds since powers are normalized to $[0, 1]$ in our work.

Step 2: *Contraction to the function class.* Since the loss is $L_{\mathcal{L}}$-Lipschitz, the contraction inequality for Rademacher complexity [5, Lemma 5.7] yields

$$\mathcal{R}_N(\mathcal{L} \circ \mathcal{F}) \leq L_{\mathcal{L}} \mathcal{R}_N(\mathcal{F}). \tag{27}$$

Therefore,

$$\mathcal{E}_{\text{est}} \leq L_{\mathcal{L}} \mathcal{R}_N(\mathcal{F}) + \sqrt{\frac{8 \ln(2/\delta)}{S}}. \tag{28}$$

By direct computation, the Lipschitz constant, for MSE loss with arguments $a, b \in [0, 1]^{2K}$, becomes $L_{\mathcal{L}} = ||\nabla_a \ell(a, b)|| = \frac{\sqrt{2}}{\sqrt{K}}$. Hence

$$\mathcal{E}_{\text{est}} \leq \frac{\sqrt{2}}{\sqrt{K}} \mathcal{R}_N(\mathcal{F}) + \sqrt{\frac{8 \ln(2/\delta)}{S}}. \tag{29}$$

Step 3: *Bounding $\mathcal{R}_N(\mathcal{F})$ for neural networks.* We rely on the idea of norm-based Rademacher complexity bounds for neural networks [6]. The TT is a composition of D layers of affine transformations and ReLU activations. Denote the weight matrices of these layers as $W_1, \ldots, W_D$, and let $\|W_j\|_F$ denote the Frobenius norm of the j-th weight matrix. The input domain is bounded such that its ℓ_2 norm satisfies $\|X\| \leq B_X$ with $B_X = \sqrt{2(K + L)}$ since AP/UE positions are normalized to the range $[0, 1]$. By Theorem 1 and [6, Remark 1], the Rademacher complexity satisfies

$$\mathcal{R}_N(\mathcal{F}) \leq \frac{B_X \left(\sqrt{2 \log(2)\, D} + 1\right) \prod_{j=1}^{D} \|W_j\|_F}{\sqrt{S}}. \tag{30}$$

Step 4: *Decomposition of the Tree-Transformer architecture.* In the proposed architecture, the depth can be decomposed as

$$D = D_0 + \lceil \log_2 K \rceil, \tag{31}$$

where D_0 corresponds to the embedding, Transformer and decoder blocks. Due to the W_{merge} weight sharing in the binary tree, the product of Frobenius norms satisfies

$$\prod_{j=1}^{D} \|W_j\|_F = C_0 \cdot \|W_{\text{merge}}\|_F^{\lceil \log_2 K \rceil}, \tag{32}$$

where $C_0 = \prod_{j \in \text{non-Tree}} \|W_j\|_F$ is a constant independent of K that depends only on the non-Tree part.

Since the model is trained with AdamW, which applies weight decay and thus penalizes large weight norms during training to encourage weights with controlled norm, we can assume $\|W_{\text{merge}}\|_F \leq C_1$ for some constant C_1. Then, we get

$$\prod_{j=1}^{D} \|W_j\|_F \leq C_0 \cdot C_1^{\lceil \log_2 K \rceil}, \tag{33}$$

Remark: In practice, C_1 value is controlled during training to avoid its growth. Particularly, AdamW weight decay training encourages $C_1 \leq 1$ by controlling $\|W_{\text{merge}}\|_F$. Moreover, the constant C_1 can be even enforced to 1 during training by using Projected Gradient Descent onto the Frobenius norm ball.

Step 5: *Final estimation error bound.* Combining (33) and (30), the Rademacher complexity bound becomes

$$\mathcal{R}_N(\mathcal{F}) \leq \frac{B_X \left(\sqrt{2\log(2)\, D} + 1\right) C_0 \cdot C_1^{\lceil \log_2 K \rceil}}{\sqrt{S}}. \tag{34}$$

Substituting into (29) yields

$$\mathcal{E}_{\text{est}} \leq \frac{2\sqrt{1 + \frac{L}{K}} \left(\sqrt{2\log(2)\, D} + 1\right) C_0 \cdot C_1^{\lceil \log_2 K \rceil}}{\sqrt{S}} + \sqrt{\frac{8\ln(2/\delta)}{S}}, \tag{35}$$

Proof of Theorem 3

To prove Theorem 3, we apply the following steps:

Step 1: *Bounding the per-UE spectral efficiency (SE) gap.* By Assumption 2, for each user $k = 1, \ldots, K$,

$$|\text{SE}_k(p^\star) - \text{SE}_k(\hat{p})| \leq L_{\text{SE}} \|p^\star - \hat{p}\|_2. \tag{36}$$

Taking expectation with respect to X,

$$\mathbb{E}_X \left[|\text{SE}_k(p^\star) - \text{SE}_k(\hat{p})|\right] \leq L_{\text{SE}}\, \mathbb{E}_X \left[\|f_{\hat{\theta}}(X) - f^\star(X)\|_2\right]. \tag{37}$$

Using the Cauchy-Schwarz inequality

$$\mathbb{E}_X \left[\|f_{\hat{\theta}}(X) - f^\star(X)\|_2\right] \leq \sqrt{\mathbb{E}_X \|f_{\hat{\theta}}(X) - f^\star(X)\|_2^2}. \tag{38}$$

Combining the above results,

$$\mathbb{E}_X \left[|\text{SE}_k(p^\star) - \text{SE}_k(\hat{p})|\right] \leq L_{\text{SE}} \sqrt{\mathbb{E}_X \|f_{\hat{\theta}}(X) - f^\star(X)\|_2^2}. \tag{39}$$

The power prediction error satisfies

$$\mathbb{E}_X \|f_{\hat{\theta}}(X) - f^\star(X)\|_2^2 = \mathcal{E}_{\text{approx}} + \mathcal{E}_{\text{est}}. \tag{40}$$

Substituting (40) into (39) gives

$$\mathbb{E}_X \left[|\text{SE}_k(p^\star) - \text{SE}_k(\hat{p})|\right] \leq L_{\text{SE}} \sqrt{\mathcal{E}_{\text{approx}} + \mathcal{E}_{\text{est}}}. \tag{41}$$

Step 2: *Bounding L_{SE}.* We derive an upper bound on L_{SE} for the uplink (UL) case. The downlink (DL) case follows analogously.

The UL spectral efficiency for user k is

$$\text{SE}_k^{\text{UL}}(p^{\text{UL}}) = \frac{\tau_u}{\tau_c} \log_2\!\left(1 + \text{SINR}_k^{\text{UL}}(p^{\text{UL}})\right), \tag{42}$$

where

$$\text{SINR}_k^{\text{UL}}(p^{\text{UL}}) = \frac{p_k^{\text{UL}} A_{kk}}{\sum_{i \neq k} p_i^{\text{UL}} A_{ki} + \sigma^2 B_k}, \tag{43}$$

with $A_{kk} = |\mathbb{E}\{v_k^H h_k\}|^2$, $A_{ki} = \mathbb{E}\{|v_k^H h_i|^2\}$, and $B_k = \mathbb{E}\{\|v_k\|^2\}$, which depend only on channel statistics. Note that for all $p^{\text{UL}} \in [0, \bar{P}^{\text{UL}}]^K$,

$$\sum_{i \neq k} p_i^{\text{UL}} A_{ki} + \sigma^2 B_k \geq \sigma^2 B_k > 0, \tag{44}$$

which ensures that the signal-to-interference-plus-noise ratio (SINR) and its derivatives are well-defined and bounded on a compact domain.

Since the domain is compact and the gradient is bounded, the Lipschitz constant is finite. By the mean value theorem [7], the Lipschitz constant is bounded by

$$L_{\text{SE}_k}^{\text{UL}} \leq \sup_{p^{\text{UL}} \in [0, \bar{P}^{\text{UL}}]^K} \left\|\nabla_{p^{\text{UL}}} \text{SE}_k^{\text{UL}}\right\|_2. \tag{45}$$

Define

$$S_k = \sum_{i=1}^{K} p_i^{\text{UL}} A_{ki} + \sigma^2 B_k, \tag{46}$$

$$I_k = \sum_{i \neq k} p_i^{\text{UL}} A_{ki} + \sigma^2 B_k. \tag{47}$$

Then

$$\text{SINR}_k^{\text{UL}} = \frac{p_k^{\text{UL}} A_{kk}}{I_k}. \tag{48}$$

For $j = k$:

$$\frac{\partial\, \text{SINR}_k^{\text{UL}}}{\partial p_k^{\text{UL}}} = \frac{A_{kk}}{I_k}. \tag{49}$$

For $j \neq k$:

$$\frac{\partial\, \text{SINR}_k^{\text{UL}}}{\partial p_j^{\text{UL}}} = -\frac{p_k^{\text{UL}} A_{kk} A_{kj}}{I_k^2}. \tag{50}$$

Using the chain rule:

$$\frac{\partial \text{SE}_k^{\text{UL}}}{\partial p_j^{\text{UL}}} = \frac{\tau_u}{\tau_c \ln 2} \cdot \frac{1}{1 + \text{SINR}_k^{\text{UL}}} \cdot \frac{\partial\, \text{SINR}_k^{\text{UL}}}{\partial p_j^{\text{UL}}}. \tag{51}$$

For $j = k$:

$$\left|\frac{\partial \mathrm{SE}_k^{\mathrm{UL}}}{\partial p_k^{\mathrm{UL}}}\right| \leq \frac{\tau_u}{\tau_c \ln 2} \cdot \frac{A_{kk}}{\sigma^2 B_k}. \tag{52}$$

For $j \neq k$:

$$\left|\frac{\partial \mathrm{SE}_k^{\mathrm{UL}}}{\partial p_j^{\mathrm{UL}}}\right| \leq \frac{\tau_u}{\tau_c \ln 2} \cdot \frac{\bar{P}^{\mathrm{UL}} A_{kk} A_{kj}}{(\sigma^2 B_k)^2}. \tag{53}$$

Using $\|v\|_2 \leq \sqrt{K}\|v\|_\infty$, we obtain

$$\left\|\nabla_{p^{\mathrm{UL}}} \mathrm{SE}_k^{\mathrm{UL}}\right\|_2 \leq \sqrt{K} \max_j \left|\frac{\partial \mathrm{SE}_k^{\mathrm{UL}}}{\partial p_j^{\mathrm{UL}}}\right|. \tag{54}$$

Define

$$G_k^{\mathrm{UL}} = \max\left\{\frac{A_{kk}}{\sigma^2 B_k},\ \max_{i \neq k} \frac{\bar{P}^{\mathrm{UL}} A_{kk} A_{ki}}{(\sigma^2 B_k)^2}\right\}. \tag{55}$$

Then

$$L_{\mathrm{SE}_k}^{\mathrm{UL}} \leq \frac{\tau_u}{\tau_c \ln 2} \sqrt{K}\, G_k^{\mathrm{UL}}. \tag{56}$$

Taking the maximum over all users:

$$L_{\mathrm{SE}}^{\mathrm{UL}} \leq \sqrt{K} \frac{\tau_u}{\tau_c \ln 2} \max_k G_k^{\mathrm{UL}} \tag{57}$$

An analogous derivation for the DL case yields

$$L_{\mathrm{SE}}^{\mathrm{DL}} \leq \sqrt{K} \frac{\tau_d}{\tau_c \ln 2} \max_k G_k^{\mathrm{DL}}, \tag{58}$$

where G_k^{DL} is defined using the DL SINR expressions.

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